

Error :- events that interrupts the normal flow of execution of program are called as errors

↓
two types

Compile time error

= occurs at compile time

ex: `char c = "hello";`

↓
run time errors

- occurs at runtime

ex: opening a file
opening db connection
dividing 2 nos by taking user input

"A run time errors that can be cured or handled are called as exception"

ex: preventing from entering 2nd no as zero for division

"the process by which we cure exception is

called as exception handling"

"exception handling is a process that provides alternate ways for normal execution of program"

Risky code:

the code which might raise an exception is called as risky code → inside try block

handling code:

the code which we used to handle the respective risky code is called as handling code → inside catch block

```
try {  
    risky code;  
}  
catch (class-name) {  
    handling code;  
}
```

→ in java there is
dedicated class
for every exception

Exception is a
parent class of all
exceptions

- if any exception raised inside try (only) block, then JVM creates its object & it looks for respective class in catch block. therefore we write class name with catch block.
- if there are multiple exception inside ^{one} try block then there can be multiple catch block.

line 1

try

{

line 2

line 3

{

catch class

{

line 4

}

line 5

exception raised at line	handled?	lines executes	termination
line 1	x	x	ab
line 2	x	1	ab
line 2	✓	1, 4, 5	n
line 3	x	1, 2	ab
line 3	✓	1, 2, 4, 5	n
line 4	x	1, [2]	ab
lines	x	1, [2, 3, 4]	ab

Conclusion:-

1. unhandled exceptions are always leads to ab termination
2. line on which we have exception, never get executed
3. exception raised outside a try block is always ab termⁿ (handled by JVM)
4. in try block, code after line on which exception is raised, will ~~not~~ execute.
never get